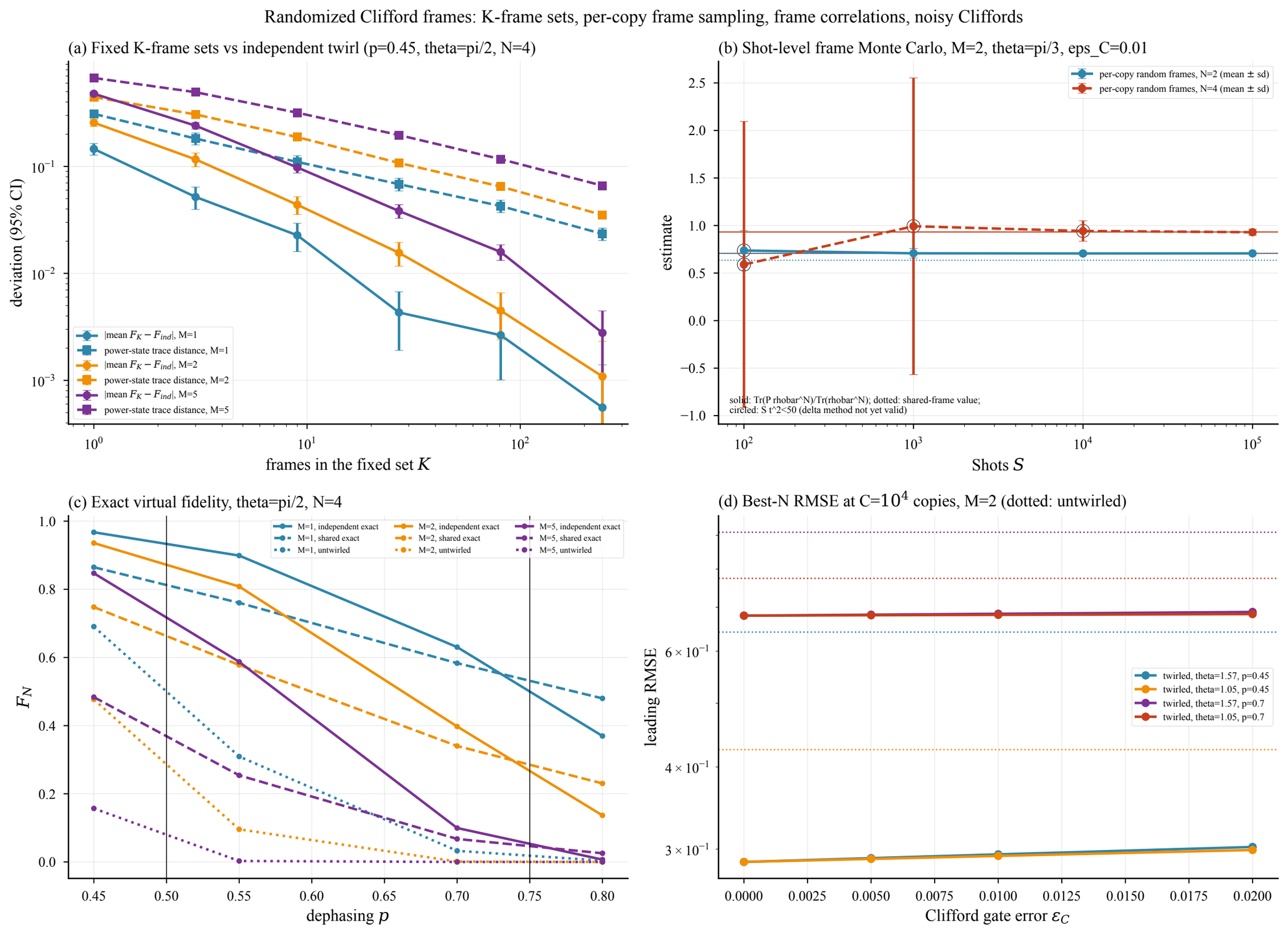

Randomized Clifford frames: K-frame sets, per-copy frame sampling, frame correlations, noisy Cliffords
(a) Fixed K-frame sets vs independent twirl (p=0.45, theta=pi/2, N=4)
deviation (95% CI)
frames in the fixed set K
|mean $F_K - F_{ind}$|, M=1
power-state trace distance, M=1
|mean $F_K - F_{ind}$|, M=2
power-state trace distance, M=2
|mean $F_K - F_{ind}$|, M=5
power-state trace distance, M=5
(b) Shot-level frame Monte Carlo, M=2, theta=pi/3, eps_C=0.01
estimate
Shots S
per-copy random frames, N=2 (mean ± sd)
per-copy random frames, N=4 (mean ± sd)
solid: Tr(P rhobar^N)/Tr(rhobar^N); dotted: shared-frame value;
circled: S t^2<50 (delta method not yet valid)
(c) Exact virtual fidelity, theta=pi/2, N=4
F_N
dephasing p
M=1, independent exact
M=1, shared exact
M=1, untwirled
M=2, independent exact
M=2, shared exact
M=2, untwirled
M=5, independent exact
M=5, shared exact
M=5, untwirled
(d) Best-N RMSE at C=10^4 copies, M=2 (dotted: untwirled)
leading RMSE
Clifford gate error ε_C
twirled, theta=1.57, p=0.45
twirled, theta=1.05, p=0.45
twirled, theta=1.57, p=0.7
twirled, theta=1.05, p=0.7